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PHY 122 — Past Questions & Materials
by Renance

Course: PHY 122
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 3
Levels: 100

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Paper 1 of 3 — PHY 122 - 2022/2023 - Second Semester - Astronomy

Level 100 • Second Semester • PastQuestion

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Paper 2 of 3 — PHY 122 PQ answer corrected

Level 100 • First Semester • PastQuestion

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4. Convert the following to Celsius

Cand-17°

3. a) Why is the temperature of the sun's surface 25°C ;

© temperature used in the ideal gas law, Celsius (b), Fahrenheit (d) any of the preceding

5. (A) constant (B) increase (C) decrease (D) temperature

* How many moles are there in 40g of water

The air in a balloon of volume 0.10 m³ &

7. What is the volume occupied by

2B The density of mercury is 13.6 x 10³ kg/m³ at 0°

712,

13.

3 hours a calorimetry experiment, 0.5

16.

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a

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6°25.

' temperature is constant

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~ \$3 (b) 1.5m (c) 2.0 m (d) 0.1294

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SOF MNTÉQUE.OL AteU gas are

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TIME; 30MINS.

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Ave ASO, i

40 Cand-29" CUB) 535° Cand-29°

he temperature of a quantity of gas, is decreased, "How is the density affected if the pressure is held constant? Increase and decrease each 4,

(A) molar)? Sais a

0.101 a pressure 0 aH if the volume of the balloon increases

temperature, what is the pressure?(A) 1.3 X: patsy !2 x10 AGG So Pa (D) 1.6 10° Pa ;

Og bf hydrogen Under a pressure of, 7,00 atm. and a temperature of 300K2(Al0.0247m? (B) 0.227m (C)

EC al GIST PIGsSure fon 2/OC0 qU/UL, Calculer rf - B

cc §

13.4% 10? ke/m? (BY 1.36x 10? kg/m? (C) 1.34 x10 j

YD .

es (0 :|izm? at a constant

al

0.247m3 (1D) 0.02271n?

'(of linear expansion are (a) ni

The units of the thermal coefficient

kg/m? (1D) 1.56 x 10? ke/m? nu le frac

If the kinetic energy of air average ideal gas molecule

313°C (d) none of the preceding?2 '05 JF i2) 2.22608 IPE 1d) 2

What is the average kinetic energy' per'molecule' of an ideal gas' at a temperature

9-747 of \ ows: (

average (rms) speed of!

(D) 1.77x103m/s j

The SI unit of heat energy is the

calorimeter cup. The cup has a'

'of the metal?(A) 41x 107//(leg- cy)

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change in the temperature of the

mass of 0.250kg, (a) if the final te

(Hi)2.2x10°)/(kg.¢°) (C)) 3.6x10°F/ (ed: é

'A' person shakes a sealed, insulated bottle containing coffee for a fe

large decrease (b) a slight decrease (fo'change (d).a ine)

'A container holds 0,50 m3 of oxygen at an absolute pressure of 4.0 atm

volume of the gas until the pressure drops to 0 atm, If the temperature

minutes, W

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Avalvgis 'openddd, allowing the gas to drive a pist

constant, what new volume does the gas 0

d the temperature outsi

recile (a) $1.3 \times 10^{\circ}W$

1 U6Pa at $25.0^{\circ}C$ is allowed to expand to'

A wali mad@ af Wood 4.00 om. thick

is thermal cnergy transported {hrough the wall by condu;

One way to cool a gas is to let i pure. Wilen a Certain fb. (

urij,inal volume, its final pressure js $1.07 \times 10^{\circ}$ PaWhat is its final temper ature?(a)

$177^{\circ}C$ (b)233K(c)212K (Q191 K

Find the root-mean-square speed 'of a'metliaite' gas molecule ($CHi4y'$ at $25.0^{\circ}C$.) 515 uals (b) 681
avs (c) 724 m/s (d) 428 mils i

How much energy is required'to raise the temperature of 5.00 ky of ledd from $20,0^{\circ}C$ to ils i /slting
point of $327^{\circ}C$? (a) $404 \times 10^{\circ}1$

(b) $8.15 \times L0^{\circ}3$ (c) 2.1321 047 (q). $1.96 \times 10^{\circ}F$ -

Two tholes of an ideal gas eated at constant pr

(A)1310N(B)1341(QQ133 0s(D)3000F

A container contains about 22! moles of air. If it is c

at constant volume?(A) 3.333 lJ/mol K(b) 2.79)/mol K (c) 4.532 Smo!

what is the change in internal energy? i

728% $105JEr$ (b) $2.32 * 105 JPr$ (c) $-2.22, \ll 105 Ic$ (@) $2.32 \ll 105 JFr \sim$

ic-ideal gas and the yolume increase while the presswie is kept consti

(yye/s(e) $3/5$ (a) $3/4$ en

cess is one in which the volume is constant(B) An adiabatic process is one witht

a is a constant-pressure process(D) An isobaric process is one in which the 3

7Y

essure from $270C$ to $1070C$, Calcilate.the work done by the gas

eee

ooled from $40^{\circ}C$ to $12^{\circ}C$ at 1,00 atm. ideally, what is the molar specific heat capacity;

1K (d) $2.920, J/mol K$

Pleat Q fiéws into a monoatom

used to do the expansion work of the gas? (a)2/d

Which onc'of the following is correct (4) An isochoric pro

heat transfer'into or out of a systern(Cy An isothermal oes

: 7

thermal equilibrium with each other. Thi

if'a syotetrn'A and B are each in thermal equilibrium with' third udy C, then A and B dre also in

statement is known as (A) First law of thermodynamics (3) Second law of thermodynamics (C) Third law of thermodynamics see ray Ry

A monoatomic ideal gas expands slowly to twice its original volume doing 300 J of work in the process. The change in the internal energy of the gas if the process is isobaric (a) $Q = 300$ J, $\Delta V = 0$ (b) $Q = 750$ J, $\Delta V = 0$ (c) $Q = 450$ J, $\Delta V = 0$ (d) $Q = 900$ J, $\Delta V = 0$ (e) $Q = 450$ J, $\Delta V = 0$

$\Delta V = 750$ J (d) $Q = 900$ J, $\Delta V = 0$ See

If a body of mass 85 kg takes in 350 J of energy;

1760: (c) 411 m (d) 420.2 m.

The amount of heat required to raise the temperature

of an ideal gas to heat it from 100°C to 250°C at constant pressure, the gas does 223 J of work

experiment adds 970 J of heat to 1.75 mol of an

ideal gas. Calculate the change in internal energy of the gas. (a) 470 J (b) 390 J (c) 0 J (d) 970 J

(e) 0 J (f) 970 J (g) 0 J (h) 970 J (i) 0 J (j) 970 J

"How high will the burn off the energy. ($V_k \text{ cal} = 1000 \text{ cal} = 4190$)"

temperature of a 1 kg body through 10°C is 30. (a) thermal heat (b) heat gained (c) thermal capacity (d) thermal energy

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heat transfer from the body, with

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slag } thermal equilibrium (D) heat energy | y/i

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"Academic Session' at 2015/2016) 1.400. Bind the fonap: Time Allowed: 2 hrs.

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TAS quantity: Gf, heat, transferred into the molecules of an object

so coldness of a body or mean Kinetic energy of molecules of a body
 energy of the individual molecule

Temperature is a measure of,
 measure of the degrees of freedom

8 they are in therm

The Zeroth law: determines
 determine temperature change

-23, A motor tyre is inflated to a pressure of 2.0×10^5 Pa

25. The average rate of heat flow in a

W

Boyle's law and Charles's

All gases at low pressure obey

the behavior of

Y ideal gases obey

18. Identify one of the statements below that is NOT correct, and state which
 law perfectly describes it. There is no real gas in practice. Real gases behave as an ideal gas at
 high temperature and low pressure and low temperature and high pressure.

an ideal gas. Diffusion of gases?

19- Which of the statements below DO NOT describe Graham's law of effusion?

Since the rate of effusion is proportional to the square root of the molecular mass, the rate of effusion is

inversely proportional to the square root of its density.

inversely proportional to the square root of its density, the rate of effusion is

20. At constant pressure the density of a fixed mass of gas is proportional to its temperature.
 proportional to its temperature.

proportional to its temperature.

21. In a gas experiment; the pressure of the gas is plotted against its volume.

a graph is obtained. The unit of the slope of the resulting curve is
 D. energy/m?

22. A gas occupies a volume of 300 cm³ at a temperature of 300 K.
 temperature is constant? A, 150 K? B, 273 K? C, 300 K? D, 450 K?

'be the pressure in it at 87°C; assuming that the volume of the tyre is kept constant? A, 2.6

2.6 x 10⁵ Pa? B, 0.2 x 10⁵ Pa? C, 1.3 x 10⁵ Pa? D, 1.3 x 10⁶ Pa?

new pressure. A, 6.85 Pa? B, 0.86 Pa?

kinetic theory of gases

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® - : student to raise the temperature of

Mash SHOT Ylogg FY, and

/ ' S 2 the work done on the sphere decreases as the radius increases

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te Laat is tl sowcild heat vay ty? A hz) RO ; Y te

37. 'The ineorcase in volume per unit volume per unit temper ature|rise when the liquid is heated. in an i

expandable containeryistealled? OY Apparent cubic Leh ansivity Be real cubic expansivity. C. linear :

expansivity D. volume oxpansivityiged dicing dae ebotinige? A,O Ae

38. Which of the following ap plicatio Ss 3348 thermal © xpansion considered.a disadvantage? A, fire alarms

"4 B. bimetallic strips of thermogin honyeetion cur it of liquid Sept in'roads,and railways, tracks D, na rivets of ship hulls.o!lowi PENCE Hap aba beet 2 a8 ley ay

ba 39° The ST unit of lincat expansivity isigi en as? AK seh 'Cy Keomnk! Dz 'Sulet, i

tz 40. The amount of heat required tolraige thétemperature! ofialbody. by tk (or 1 °C) is called? A. thermal

os hex at{BPHeat capacity Ce Spevitic heat capacity; D. linearjexpansivity. fx :

a 41. Tf 1008 of water requires 12600J of heatito,ra temperature from, 30°C to CRA. ae i

af capacity? A. 342 TRYB)4204 TRS F445 40 TK * iy

' 42. The quantity of heat required sto'K ise the toniperature ofl: kg mass ofa substance bys.

ie called? A, heat energ QOspecific heat capiteity,C. heat eapacity.D.-Calories. 5). 5:

f 43. In an experiment to determine-the specitic.heat capacity. ofa solid material , which ofthe, following

' . apparatus,is NOT required to caity;ioul he experinient? Ae ee glectric heat -ostat.

j i : 44. Row much heat is requited to raise 'tlic teniperature ot 0. kg. of water by, LOC, ifthe hi

: water ig 420. TTK) 4200 F-BusS 1000 JD1:3200,Taliener

5

45) Wheii heat is supplied

sphere increases By

internal energy of the sphet }

46. When a 0.24kg brass: cylinderiis. heated using a 1 20005, heater, its | eomperatars increases from 30 °C to

IC! B. 169.1 kg! pee Teg?

f : 100 °C in 3.2.s. Whatis the specifichieat cane

| D. 400 I ket Ks :

I j ; cup of Lipton:

| : : 2000, JK. how.

| ae "100 W. what is th tise in temperature

' ag vv hich one of the following i is-'NO

i yf.the,

oe path C. root'mean .

lisions is know? as? A. wavelength {3} mean free poe an

56, The distance two, successive col

E square velleciy D; none oi the above. :

57, A vessel contain hydrogen gas at 0 °C and a pressure of s a.V0)

number of moles of hydrogen in the vessel. Take Rea Jimol.-K, yee

~~~ mole-of hydrogen = 28g. A. 5,30 x10? mole B. 4,53 x10 mole C, 6.7%) U,

58. Calculate the root-mean- square speed of an alr molecules at a temperature

mole of air = 29g and Avogadro's number = 6,023 x.108, A, 300 sn/s B.2

um at a temperature of 550K.

m/s. A

A: 1.04 10° JB. 3.20 x 10°

\$ Cale

¢ of 60 atm has a vole ei Oa ances een ©

mole D: 3.50 x10 mole, i

of 25 oe-Take mass of one---- \*

57 m/s C. 530 m/s D. 250

xygen in thermal equilibri

59, A container holds a mixture of hydrogen and 0

f both types of molecule.

what is the mean translational kinetic energies 0

PT C:2.43,x 1022F D.3.00 x 1022 Toiicts

60, 'The value of Boltzmatin's constant is? AL13.% Os Ss 4

E "61. ...1s. Describe the gross characteristics of jarge'scale properties which sugecs' ed more or less: directly

by our sense of perception? A, Process B. microscopic system,C, macroscopr® system D, surroundings.

; { atl e gas is, compressed

62, A gas-has'a 'volume of.80.m' at STP: Calculate the new. temperature, if the "ga

PS', take Cp'2 1.68 KI kg! Ke! and Cv.=)1.20 KKB: AAT ECR.

KAD, 8,314 5K"

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Course Title. - . Heat and Kinetic Theory - are

“Course Coder». PHY 122iTesh—At yo “4

Instructions: ~ \_ ANSWER ALL. THREE’ UESTIONS ©

\_ MQuestion qh ae,

\* A) What is thermal Equillsurt ay Explain the.Zeroth law.of thermodynamics = ve

B) i) At what temperature does the Fahrenheit qnd Celsius Temperature sea eh ;

: the same value, Ve. ae, — 7 :

ii) List three types of thermometers yout | nol iy and. state

« iN. 5

| -° ~ disadvantages of each, .

‘two advantages and

Question 2

A)Lis

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C2-c--> where all saraineters take their usual meaning:

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Question 3 : Li

A) State four basic assumptions hE the kinetié theory of gas Se5,-

B) eo that the pressure in a gas contained ing a eubig box i is

P= \*YamnvPusing ure Ideal'gas. model. oe

Be) pe

## Paper 3 of 3 — PHY 122

Level 100 • Second Semester • PastQuestion

This source paper is a scanned document; its pages are reproduced below.

1. Temperature is a measure of the? A. quantity of heat transferred into the molecules of an object B. measure of the degree of hotness or coldness of a body or mean kinetic energy of molecules of a body C. amount of work done by the molecules of the object D. kinetic energy of the individual molecules of the substance.
2. Which of the following is NOT necessary true about two bodies in thermal contact? A. Heat flow within them B. there is heat transfer from a body with higher temperature to one with lower temperature C. they are in thermal equilibrium D. heat energy transfer takes place.
3. The Zeroth law of thermodynamics has its useful application in A. Design of a thermometer B. determine temperature change C. determine upper fixed point D. all of the above
4. The interval between the upper fixed point and the lower fixed point of a thermometer is called? A. Empirical scale B. Rankle scale C. thermometric property D. fundamental interval.
5. The temperature at which water, ice and steam are in thermal equilibrium is called? A. Absolute zero temperature B. triple point temperature C. ideal temperature D. the kelvin temperature.
6. If a real gas vaporized at the temperature of  $118^{\circ}\text{C}$ . What is this temperature on the Fahrenheit scale? A.  $125^{\circ}\text{F}$  B.  $300.3^{\circ}\text{F}$  C.  $244.4^{\circ}\text{F}$  D.  $325^{\circ}\text{F}$ .
7. A solid oxygen melt at temperature of  $-218.4^{\circ}\text{C}$ . Find the temperature of this gas on the Kelvin scale? A.  $67\text{K}$  B.  $176.43\text{K}$  C.  $72.21\text{K}$  D.  $54.76\text{K}$ .
8. At what temperature does the Fahrenheit and Celsius temperature scales have the same value? A.  $-40$  B.  $-20$  C.  $20$  D.  $40$
9. A well lagged bar of length  $100\text{ cm}$  has its ends maintained at  $100^{\circ}\text{C}$  and  $40^{\circ}\text{C}$  respectively. What is the temperature at a point  $60\text{ cm}$  from the hotter end? A.  $58^{\circ}\text{C}$  B.  $62^{\circ}\text{C}$  C.  $64^{\circ}\text{C}$  D.  $76^{\circ}\text{C}$
10. Which of the following is NOT an advantage of a Mercury-in-glass thermometer? A. it is suitable for normal use B. it is bulky C. easy to use D. it is cheap to buy and portable.
11. Which of the following is NOT a good requirement and characteristics of a thermometer? A. Sensitivity B. Reproducibility C. speed D. high thermal capacity.
12. The temperature in the Antarctic during the mid-winters is found to be  $212\text{ K}$ . What is this temperature in  $^{\circ}\text{C}$ ? A.  $-54\text{ K}$  B.  $76\text{ K}$  C.  $-61\text{ K}$  D.  $485\text{ K}$ .
13. A brass rod is  $2.00\text{ m}$  long at a certain temperature. What is its length for a temperature rise of  $100\text{ K}$ , if the linear expansivity of brass is  $19.0 \times 10^{-6}\text{ K}^{-1}$ ? A.  $3.00\text{ m}$  B.  $2.004\text{ m}$  C.  $2.0\text{ m}$  D.  $3.021\text{ m}$
14. Water shows anomalous behavior A. below  $0^{\circ}\text{C}$  B. between  $0^{\circ}\text{C}$  and  $4^{\circ}\text{C}$  C. at exactly  $4^{\circ}\text{C}$  D. between  $4^{\circ}\text{C}$  and  $100^{\circ}\text{C}$
15. The linear expansivity of brass is  $2 \times 10^{-5}\text{ C}^{-1}$ . If the volume of a piece of brass is  $15.00\text{ cm}^3$  at  $0^{\circ}\text{C}$ . What is the volume at  $100^{\circ}\text{C}$ ? A.  $16.03\text{ cm}^3$  B.  $16.00\text{ cm}^3$  C.  $15.09\text{ cm}^3$  D.  $14.03\text{ cm}^3$
16. A metal rod is  $5.00\text{ m}$  long at  $30.0^{\circ}\text{C}$ . If its linear expansivity is  $12 \times 10^{-4}\text{ C}^{-1}$ , to what temperature would it have to be heated in order to gain an extension of  $0.003\text{ m}$ ? A.  $60^{\circ}\text{C}$  B.  $70^{\circ}\text{C}$  C.  $80^{\circ}\text{C}$  D.  $90^{\circ}\text{C}$ .
17. A cube made of a metal of linear expansivity  $\alpha$  is heated to a temperature  $\theta$ . If the initial volume of the cube is  $V_0$ , the correct expression for the increase in volume of the cube is? A.  $\frac{1}{3}\alpha V_0\theta$  B.  $\frac{1}{2}\alpha V_0\theta$  C.  $2\alpha V_0\theta$  D.  $3\alpha V_0\theta$

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### Linear Expansion of Materials

As the temperature of a material increases, it begins to expand. For instance, a metal rod or beam will increase its length by an amount  $\Delta L$ . The value of  $\Delta L$  depends on the original length ( $L_0$ ) at the original temperature ( $T_0$ ), the temperature to which it is heated ( $T$ ) and the coefficient of linear expansion ( $\alpha$ ). The equation relating these variables is shown in **Figure 1**. The amount of expansion depends upon the material; each material has its own unique coefficient of linear expansion. Coefficients for a variety of materials are shown in **Table 1**. Note that the left column includes metals and metal alloys and the right column includes non-metals.

**Figure 1**

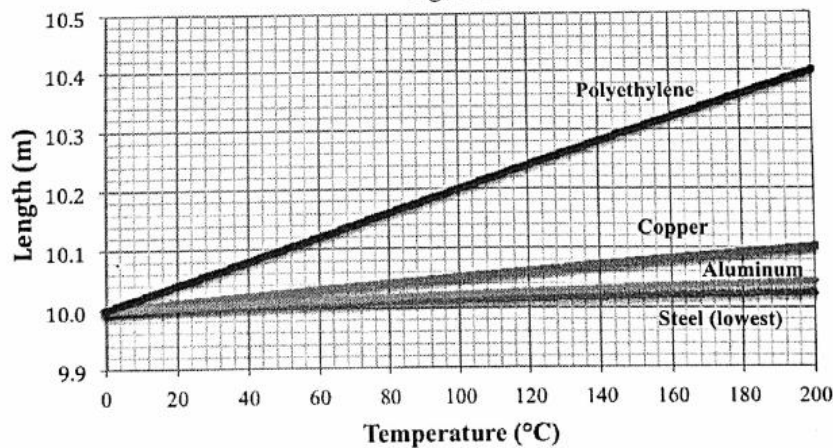
$$\Delta L = \alpha \cdot L_0 \cdot (T - T_0)$$

**Table 1**

| Material | $\alpha$ (m/°C/m)   | Material           | $\alpha$ (m/°C/m)   |
|----------|---------------------|--------------------|---------------------|
| Copper   | $5.1 \cdot 10^{-5}$ | Wood               | $5.0 \cdot 10^{-6}$ |
| Steel    | $1.3 \cdot 10^{-5}$ | Polyethylene       | $2.0 \cdot 10^{-4}$ |
| Aluminum | $2.2 \cdot 10^{-5}$ | Polystyrene        | $7.0 \cdot 10^{-5}$ |
| Brass    | $1.9 \cdot 10^{-5}$ | Polyvinyl chloride | $5.0 \cdot 10^{-5}$ |

The expansion of a rod or beam in one dimension as described by **Figure 1** is known as **linear expansion**. Linear expansion causes the length of a rod or beam to change with temperature. **Figure 2** shows how the length of a beam varies as a function of temperature for four different materials.

**Figure 2**



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**Questions:**

1. The equation in **Figure 1** would suggest that the amount of expansion ( $\Delta L$ ) of a material would be \_\_\_\_\_.
  - a. directly proportional to the temperature of the material
  - b. directly proportional to the temperature change of the material
  - c. inversely proportional to the temperature of the material
  - d. inversely proportional to the temperature change of the material
2. Which one of the following quantitative conclusions regarding the relationship between temperature and length is consistent with the equation in **Figure 1**?
  - a. The length of a rod will be twice as long at twice the temperature.
  - b. As the temperature change is doubled, the length of the metal rod doubles.
  - c. A mere doubling of temperature is enough to cause a rod to double its length.
  - d. The amount of expansion of a rod is twice as much for twice the temperature change.
3. According to **Table 1** and the **Figure 1** equation, which one of the listed materials has the lowest tendency to expand when heated?
 

|          |                 |
|----------|-----------------|
| a. Brass | b. Polyethylene |
| c. Steel | d. Wood         |
4. Use the **Figure 1** equation and the **Table 1** to determine which of the following situations would result in the greatest amount of expansion ( $\Delta L$ ).
  - a. Increase the temperature of a 10.0-m length copper rod from 20°C to 30°C.
  - b. Increase the temperature of a 5.0-m length copper rod from 120°C to 130°C.
  - c. Increase the temperature of a 10.0-m length brass rod from 20°C to 30°C.
  - d. Increase the temperature of a 5.0-m length brass rod from 120°C to 130°C.
5. Use the **Figure 1** equation and the **Table 1** to determine which of the following situation would result in the greatest amount of expansion ( $\Delta L$ ).
  - a. Increase the temperature of a 5.0-m length copper rod from 50°C to 100°C.
  - b. Increase the temperature of a 5.0-m length copper rod from 50°C to 200°C.
  - c. Increase the temperature of a 10.0-m length copper rod from 50°C to 100°C.
  - d. Increase the temperature of a 10.0-m length copper rod from 50°C to 200°C.
6. The amount that a 2-meter length copper rod expands for an increase in temperature from 25°C to 35°C is hardly noticeable. Use the **Figure 1** equation and the **Table 1** to identify the correct explanation for this observation.
  - a. The  $\alpha$  value is quite small for copper; the expansion would be much more noticeable for wood with its larger  $\alpha$  value.
  - b. The 25°C to 35°C range is low on the temperature scale; it would be more noticeable for a change from 125°C to 135°C.
  - c. The  $\alpha$  values of the materials in **Table 1** are very small; larger temperature changes are required to notice the expansion.
  - d. The expansion is very small compared to the length of the rod; it would be a more noticeable expansion when it occurs for a shorter rod.

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7. Each line in **Figure 2** is sloped with a different slope. The slopes of the lines are representative of the \_\_\_\_\_.
  - a. rate at which the beam is heated
  - b. the molar mass of the different materials
  - c. the original length of the different beams
  - d. coefficient of linear expansion of the material
8. If each of the beams in **Figure 2** were cooled to  $-50^{\circ}\text{C}$ , the shortest beam would be \_\_\_\_\_.
  - a. aluminum
  - b. copper
  - c. polyethylene
  - d. steel
9. Which description best describes the origin of the four plotted lines in **Figure 2**?
  - a. Beams of four different materials and having different lengths were obtained at  $0^{\circ}\text{C}$ . The beams were heated to  $200^{\circ}\text{C}$ ; their length was measured at various temperatures.
  - b. Beams of four different materials were obtained at  $0^{\circ}\text{C}$  and measured to be the same length. They were heated to  $200^{\circ}\text{C}$ ; their length was measured at various temperatures.
  - c. Beams of four different materials with random lengths were obtained at  $200^{\circ}\text{C}$ . The beams were cooled until they reached the same length; this occurred at a temperature of  $0^{\circ}\text{C}$ .
  - d. Four equal length beams were obtained at room temperature. They were heated and cooled between the temperature range of  $0^{\circ}\text{C}$  and  $200^{\circ}\text{C}$ ; their length was measured at various temperatures.
10. Suppose a 10.0-meter length polypropylene pipe at  $0^{\circ}\text{C}$  were heated to  $100^{\circ}\text{C}$ . According to **Figure 2**, what is the amount of expansion ( $\Delta L$ ) of the beam?
  - a. 0.30 m
  - b. 0.40 m
  - c. 10.30 m
  - d. 10.40 m
11. Suppose a 10.0-meter length polyvinyl chloride (PVC) pipe at  $0^{\circ}\text{C}$  were heated to  $50^{\circ}\text{C}$ . Based on the values in **Table 1** and the plots of **Figure 2**, what would be the amount of expansion ( $\Delta L$ ) of the PVC pipe?
  - a. Approximately 0.01 m
  - b. Approximately 0.05 m
  - c. Approximately 0.10 m
  - d. Approximately 0.50 m
12. Suppose a 10.0-m length brass beam at  $0^{\circ}\text{C}$  were heated to  $200^{\circ}\text{C}$  and its length as a function of temperature was plotted – similar to **Figure 2**. Where would the line be located on the graph?
  - a. Below the steel line.
  - b. Between the steel and the aluminum line.
  - c. Between the aluminum line and the copper line.
  - d. Between the copper line and the polypropylene line.
13. Suppose a 5.0-m length copper pipe at  $0^{\circ}\text{C}$  were heated to  $200^{\circ}\text{C}$ . Based on the equation in **Figure 1** and the plots of **Figure 2**, what would be its new length at this temperature?
  - a. Approximately 5.05 m
  - b. Approximately 10.1 m
  - c. Approximately 15.1 m
  - d. Approximately 20.2 m

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